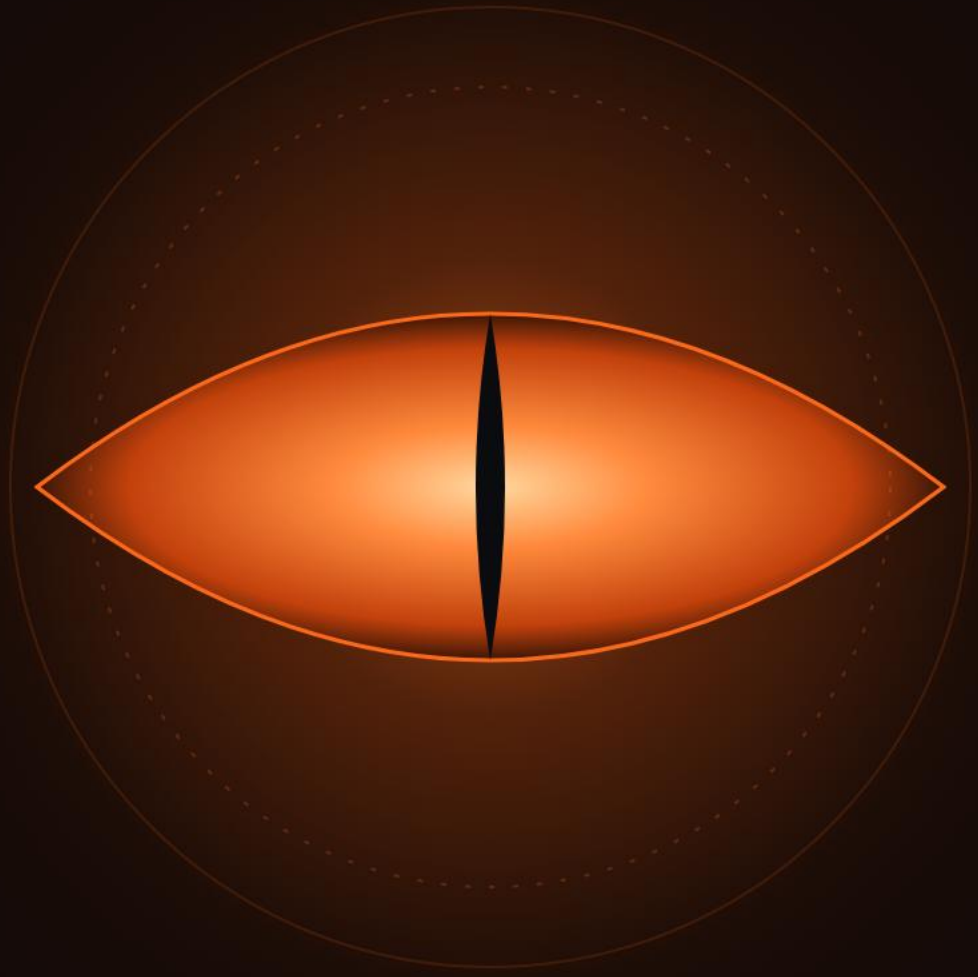
NETRA · HACKATHON · OCT 24 2026

SAURON

Autonomous site security.
It sees, it judges, it sends a drone.



THE PROBLEM

Site security doesn’t scale with people, and help arrives too late.

Guards

One post, one pair of eyes. Coverage costs more people, every night and weekend.

Remote monitoring

People watching walls of feeds. Most of what they see is nothing; the one thing that matters is easy to miss.

Response

Someone has to notice, call, and drive over. By then the lock is cut and the bike is gone.

Sauron watches every feed, decides by itself, and launches a drone.
Operators only abort.

BUILT TODAY · OCT 24

From a camera pixel to a drone launch

01

Live feeds

Axis PTZ and C920 stream to the cloud as low-latency video.

02

Person boxes

A GPU pod detects and tracks every person, drawn live on the video.

03

Sauron judge

Jev facts → policy → verdict: watch, deploy or urgent.

04

Map coordinate

A calibrated camera and sitefuse turn the person into lat/lon.

05

Send drone

Autonomous on a deploy verdict, or one click on a person box.

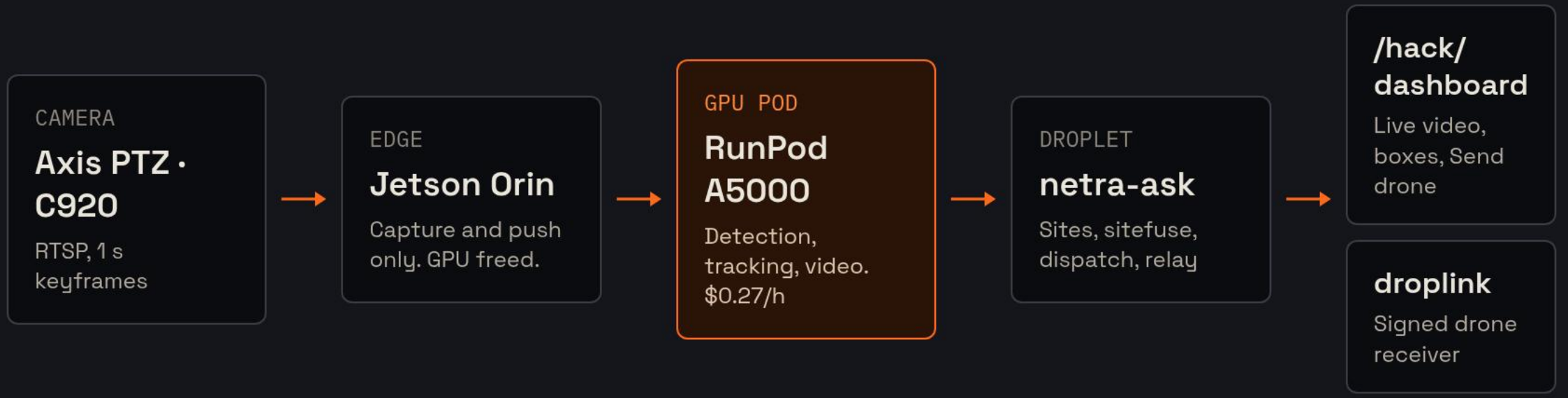
06

Signed launch

HMAC-signed incident to the drone, then a 1 Hz position stream.

ARCHITECTURE

Capture at the edge, see in the cloud

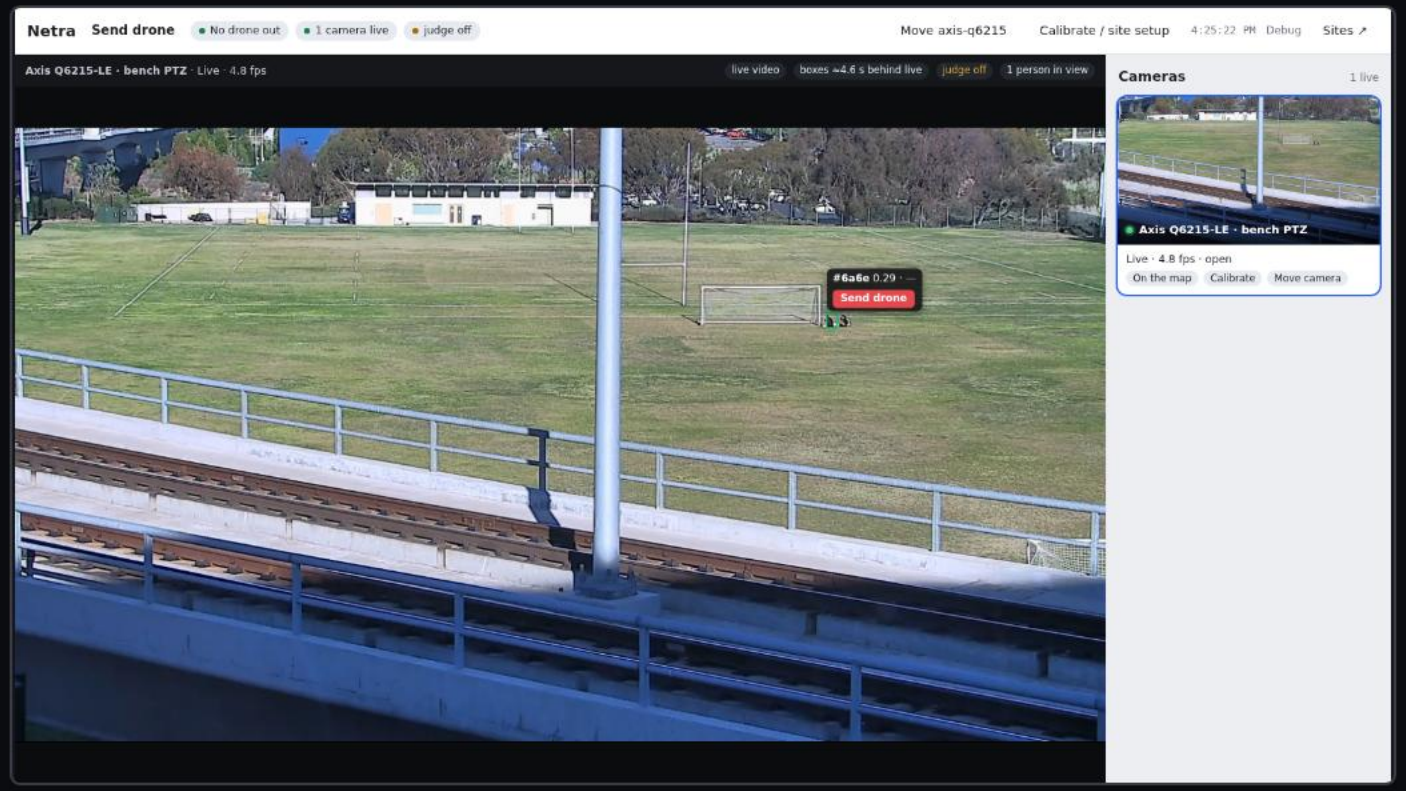


Edge → cloud, moved today. Cutover 22:19 UTC: the pod took over perception and posts to production through the relay.

Inference per frame at 5 fps: 34–46 ms on the pod vs 75–86 ms on the Jetson.

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One click from person to drone



Live video from the Axis PTZ, through the pod, in the browser.

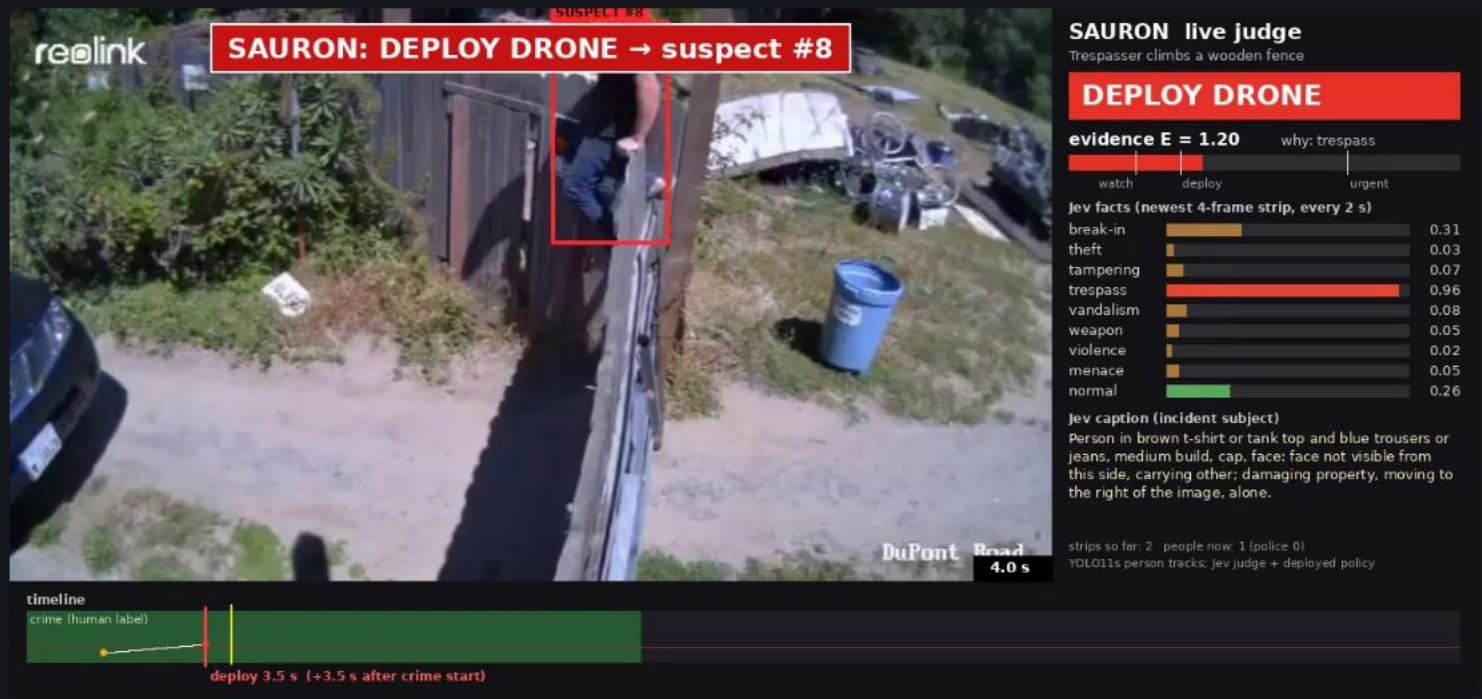
Person box with track id and confidence, and a **Send drone** button.

Honest status chips: boxes ≈4.6 s behind live, judge off, 1 person in view.

Calibrate / site setup and Move camera live in the same page.

REAL CRIME, REAL CCTV

Deploy, seconds into the crime



+3.5 s

fence climb on clean CCTV: crime start → drone deploy

5 / 5

crime clips ended in a deploy

0 / 2

benign clips deployed

EVERY CLIP

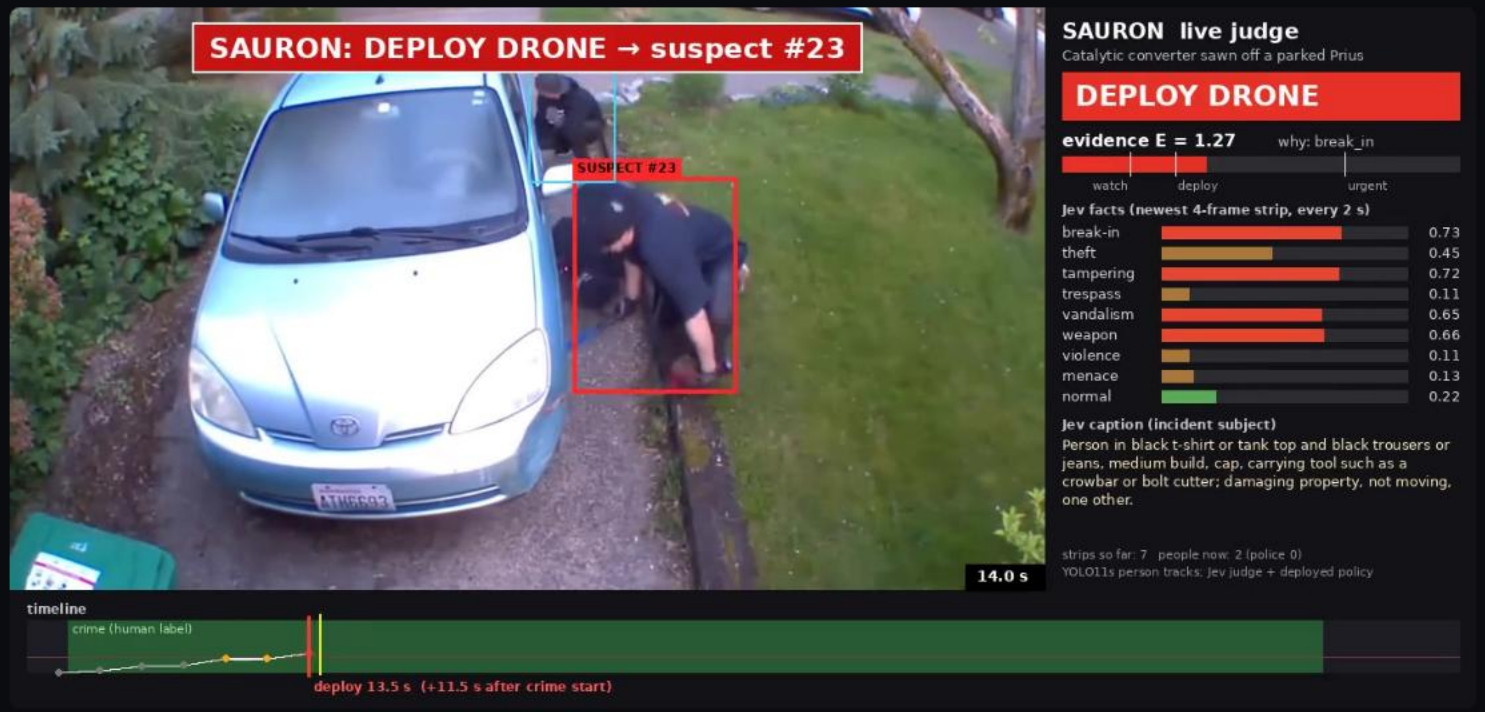
Five crimes, two owners, one policy

Clip	Crime window	Deploy at	After start	Police
Fence climb (hero)	0–12 s	3.5 s	+3.5 s	none
ATM ripped off with a chain	4–40 s	7.5 s	+3.5 s	none
Bike lock cut	2–12 s	7.5 s	+5.5 s	from 11.5 s, after deploy
Catalytic converter	2–62 s	13.5 s	+11.5 s, slow	none
Night car break-in	7–29 s	19.5 s	+12.5 s, slow	none
Owner opens own car 1	benign	never	E 0.00	—
Owner opens own car 2	benign	never	E 0.00	—

Crime windows human-labelled. Sources cleaned before the models run (news banner, app icons cropped; ATM trimmed to one camera). Police are drawn blue and never picked as the suspect. Jev cost for the whole set: $0.054.

THE JUDGE

Facts, then policy, then a verdict



1 Facts

Jev reads a 4-frame strip every 2 s and scores 9 facts: break-in, tampering, trespass, weapon…

2 Policy

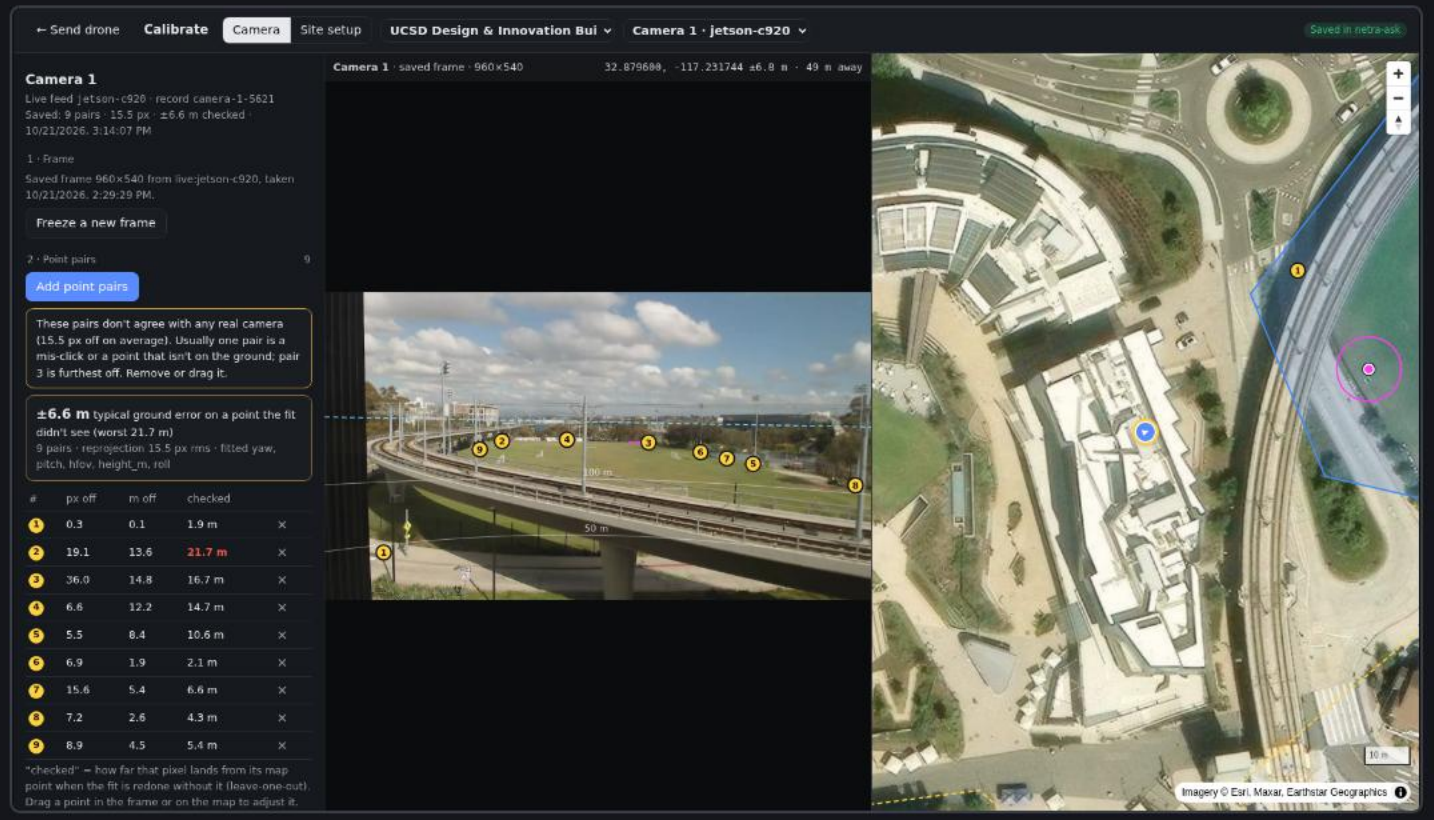
Evidence E builds up and decays (60 s half-life).

3 Verdict

Watch at E 0.6, deploy at 1.0, urgent at 2.5. Here: E 1.27 → deploy.

PERSON → COORDINATE

Pixel pairs become a camera pose



±6.6 m

median leave-one-out ground error, worst 21.7 m

9 frame ↔ map pairs, 15.5 px rms. Fit gives yaw, pitch, field of view, height, roll.

C920 at the DIB window: a desk test rig, not headline evidence.

Today's fix finds an unplaced camera from 6+ pairs; it recovered the surveyed pose (7.2 m).

MEASURED TODAY

Timings from the real system

2.78 s
click → signed launch received by droplink

0.79 s
click → launch accepted by netra-ask

~0.5 s
PTZ command, browser → camera (median)

5 fps
Axis detection on the pod, 40–43 ms per frame

0.8–1.6 s
live video cold start in the browser

2.9–6.4 s
person boxes behind live: the lag to beat

Abort is ordered: end → closed → abort, all signed; nothing open after it.

WHAT'S NEXT

Put the drone in the air

Drone integration

sauron_link launches on signed incidents but only flies to the launch point. Next: follow the 1 Hz positions (PX4 Offboard).

More cameras

Calibrate the Axis PTZ, bring node3 back, and weight evals to real fixed CCTV, day and night.

Site model

Boundary, cameras and map in one place, so every person lands on the site, not just in a frame.

Judge speed

Cut box lag (edge posts every ~3 s) and slow verdicts like +11.5 s; small local models per frame.

SAURON

Sees. Judges. Sends a drone.

Netra — Pranay (CEO), Landon (CTO), William (COO)

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